SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based

COURSE NAME: COMPUTER VISION **COURSE CODE ITA05**

COURSE OUTCOME COVERED

CO2: Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)

Assessment Tool Weightage: 20%

Scenario-Based Questions

1. **Salt-and-Pepper Noise**

A scanned document contains salt-and-pepper noise affecting readability.

(a) Explain the scenario and characteristics of this noise.

(b) Identify key issues impacting image quality.

(c) Apply an appropriate filtering technique.

(d) Justify why this method is more suitable than alternatives.

(e) Present your response clearly and logically.

2. **Object Boundary Detection**

In a medical image, boundaries between tissues need to be clearly identified.

(a) Interpret the scenario and explain the importance of boundary detection.

(b) Identify key challenges in detecting boundaries.

(c) Apply a suitable edge detection method.

(d) Justify your selection with logical reasoning.

(e) Present your explanation clearly and systematically.

3. **Simple Segmentation**

An image contains objects clearly distinguishable from the background based on intensity.

(a) Interpret the scenario and explain segmentation requirements.

(b) Identify key features enabling separation.

(c) Apply a suitable segmentation technique.

(d) Justify your decision logically.

(e) Present your response clearly.

4. **Low Brightness Image**

An image captured in low light appears very dark with poor visibility.

(a) Interpret the scenario and explain the cause of low brightness.

(b) Identify key issues affecting image visibility.

(c) Apply an appropriate enhancement technique.

(d) Justify your choice with reasoning.

(e) Present your explanation clearly.

5. **Gaussian Noise**

An image contains Gaussian noise affecting smooth regions.

(a) Interpret the scenario and describe Gaussian noise.

(b) Identify key issues affecting quality.

(c) Apply an appropriate smoothing filter.

(d) Justify your method over alternatives.

(e) Present your explanation clearly.

6. **Loss of Fine Details**

An image loses fine details after processing.

(a) Interpret the scenario and explain detail loss.

(b) Identify key issues related to filtering.

(c) Apply a suitable enhancement technique.

(d) Justify your method with reasoning.

(e) Present your explanation clearly.

7. **Edge Enhancement Requirement**

An application requires highlighting edges for feature extraction.

(a) Interpret the scenario and explain the need for edge enhancement.

(b) Identify key issues in detecting edges.

(c) Apply an appropriate method.

(d) Justify your choice logically.

(e) Present your explanation clearly.

Code of Conduct:

I M. Phanith Chandu (192425238) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: M. Phanith Chandu

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Scenario 1: Salt-and-Pepper Noise

(a) Scenario Interpretation

A scanned document affected by salt-and-pepper noise shows random black and white pixels scattered across the image. This is an impulse noise type where isolated pixels are replaced entirely with the maximum (white/255) or minimum (black/0) intensity value, unlike Gaussian noise which adds small variations to every pixel. It commonly arises from scanner sensor faults, dust on the scanning glass, or bit errors during data transmission.

(b) Key Issues Identified

- Text characters become broken or unreadable due to random bright/dark specks overlapping strokes.
- The noise is high-frequency and localized, so it can be mistaken for genuine fine detail if not handled carefully.
- Standard linear smoothing (mean filter) blurs the noise into surrounding pixels instead of removing it, degrading edge sharpness.

(c) Applied Technique

A Median Filter (typically a 3×3 neighborhood) is applied. Each pixel is replaced by the median intensity value of its surrounding neighborhood rather than the average.

(d) Justification

The median filter is a non-linear, order-statistic filter, which makes it ideal for impulse noise: extreme noisy pixel values (0 or 255) are outliers within the neighborhood and are simply excluded when the median is picked, so they do not influence the output. A mean filter, in contrast, includes these extreme values in its average and blurs them across neighboring pixels, degrading text edges. The median filter therefore removes the noise almost completely while preserving sharp character boundaries, making it clearly superior to linear smoothing for this scenario.

(e) Presentation Summary

Salt-and-pepper noise is impulse-type corruption best identified by isolated extreme-value pixels; median filtering removes it effectively while retaining document readability and edge sharpness.

Scenario 2: Object Boundary Detection

(a) Scenario Interpretation

In a medical image such as an MRI or CT scan, precise boundary detection between tissues (e.g., tumor vs. healthy tissue, organ vs. background) is critical because diagnosis, measurement, and treatment planning all depend on accurately locating where one structure ends and another begins. An incorrect or missed boundary can lead to misdiagnosis.

(b) Key Issues Identified

- Low contrast between adjacent soft tissues makes intensity transitions gradual rather than sharp.
- Medical images often contain acquisition noise (from the scanning equipment) that creates false edges.
- Boundaries may be blurred, discontinuous, or vary in thickness, making simple gradient-based detection unreliable.

(c) Applied Technique

The Canny Edge Detection algorithm is applied, involving Gaussian smoothing, gradient magnitude/direction computation (Sobel), non-maximum suppression, and hysteresis thresholding with two thresholds.

(d) Justification

Canny is preferred over simpler operators such as Sobel or Prewitt because it combines noise suppression (Gaussian smoothing) with precise localization (non-maximum suppression) and produces thin, continuous, single-pixel-wide edges. The dual-threshold hysteresis step also connects weak but relevant edges to strong ones, which is essential for tracing faint, low-contrast tissue boundaries without breaking them into fragments, giving far more reliable and clinically usable results.

(e) Presentation Summary

Boundary detection in medical imaging demands both noise robustness and precise localization; Canny edge detection satisfies both requirements and yields continuous, clinically interpretable tissue boundaries.

Scenario 3: Simple Segmentation

(a) Scenario Interpretation

Since the objects in the image have intensity values clearly distinct from the background, the segmentation requirement is to separate foreground from background purely on the basis of pixel brightness, without needing complex region-growing or clustering methods.

(b) Key Issues Identified

- A bimodal histogram — two clearly separated peaks, one for object pixels and one for background pixels.
- Minimal overlap between the intensity distributions of object and background.
- Uniform illumination across the image, so intensity differences are due to the objects themselves, not lighting variation.

(c) Applied Technique

Global thresholding using Otsu's Method is applied to automatically compute the optimal threshold value that separates the two intensity classes.

(d) Justification

Otsu's method is chosen over a manually fixed threshold because it statistically analyzes the histogram and selects the threshold that maximizes between-class variance (equivalently, minimizes within-class variance), removing guesswork and adapting automatically to the specific image. For a clearly bimodal, well-separated intensity distribution as described, this method is fast, simple to implement, and highly accurate.

(e) Presentation Summary

When objects and background differ clearly in intensity, global (Otsu) thresholding provides a simple, automatic, and accurate segmentation solution.

Scenario 4: Low Brightness Image

(a) Scenario Interpretation

An image captured in low-light conditions has most of its pixel intensity values compressed into the lower (dark) end of the 0–255 range. This happens because insufficient light reaches the sensor, resulting in low overall signal and a narrow, dark-skewed histogram.

(b) Key Issues Identified

- Poor visibility of objects and details due to compressed intensity range.
- Low overall contrast, making it hard to distinguish nearby intensity levels.
- Simple brightness addition risks pixel saturation (clipping) at 255 without truly improving contrast.

(c) Applied Technique

Histogram Equalization is applied to redistribute the image's intensity values across the full available dynamic range.

(d) Justification

Histogram equalization is preferred over a simple brightness offset because it does not just shift all pixel values uniformly — it remaps intensities based on the cumulative distribution function, stretching regions of the histogram that are densely packed and giving greater contrast exactly where the image needs it most, without requiring any manually chosen parameter. This produces a naturally enhanced, higher-contrast image usable for further analysis.

(e) Presentation Summary

Low brightness stems from a compressed, dark-skewed intensity distribution; histogram equalization enhances visibility and contrast automatically by redistributing intensities across the full range.

Scenario 5: Gaussian Noise

(a) Scenario Interpretation

Gaussian noise is a statistical noise where each pixel's intensity is altered by a random value drawn from a Gaussian (normal) distribution with zero mean. Unlike salt-and-pepper noise, it affects every pixel by a small amount rather than a few pixels drastically, producing a grainy appearance especially visible in smooth, low-texture regions.

(b) Key Issues Identified

- Smooth regions (sky, skin, flat backgrounds) show a visible grainy texture that was not originally present.
- The noise is continuous and additive across the entire image, so isolated-pixel filters (like median) are less efficient here.
- Excessive smoothing to remove the noise risks blurring genuine edges and fine details.

(c) Applied Technique

A Gaussian Smoothing Filter (Gaussian blur) is applied, convolving the image with a Gaussian kernel.

(d) Justification

The Gaussian filter is chosen over a plain mean filter because it assigns weights to neighborhood pixels according to a Gaussian distribution — pixels closer to the center contribute more, distant pixels less — which matches the statistical nature of Gaussian noise and produces smoother, more natural blurring with comparatively better edge preservation than a uniform mean filter of the same size.

(e) Presentation Summary

Gaussian noise is continuous, low-amplitude corruption best matched by Gaussian smoothing, which reduces graininess while preserving image structure better than uniform averaging.

Scenario 6: Loss of Fine Details

(a) Scenario Interpretation

Fine detail loss after processing typically occurs when a low-pass (smoothing) filter with too large a kernel, or applied too aggressively, is used to remove noise. Low-pass filtering attenuates high-frequency components of the image, and fine details/textures/edges are themselves high-frequency information, so they get suppressed along with the noise.

(b) Key Issues Identified

- Over-smoothing removes noise but also blurs edges and small structures needed for analysis.
- No mechanism exists in basic smoothing filters to distinguish 'noise' frequencies from 'detail' frequencies.
- Repeated or large-kernel filtering compounds the detail loss.

(c) Applied Technique

Unsharp Masking (a sharpening enhancement technique) is applied: a blurred version of the image is subtracted from the original to extract high-frequency detail, which is then added back with a scaling factor.

(d) Justification

Unsharp masking is justified because it directly targets the problem: it recovers the high-frequency detail that smoothing suppressed and reintroduces it in a controlled, weighted manner, rather than blindly sharpening the whole image. This restores edge and texture sharpness while still keeping the benefit of the earlier noise reduction, which a simple re-filtering approach cannot achieve.

(e) Presentation Summary

Detail loss results from aggressive low-pass filtering suppressing high-frequency content; unsharp masking restores this detail by reinjecting the extracted high-frequency component.

Scenario 7: Edge Enhancement Requirement

(a) Scenario Interpretation

Feature extraction applications (such as object recognition or shape analysis) depend on clearly defined edges, since edges represent object boundaries, corners, and contours — the structural features from which shape descriptors are computed. Enhancing edges makes these features stand out for reliable extraction.

(b) Key Issues Identified

- Edges can be faint or low-contrast, making them hard to distinguish from background variation.
- Noise in the image can create false edges or obscure genuine ones.
- Edges exist at different scales and orientations, requiring a method sensitive to gradient direction and magnitude.

(c) Applied Technique

The Sobel Operator is applied, computing horizontal and vertical intensity gradients using two 3×3 convolution kernels, then combining them into a gradient magnitude map to enhance edges.

(d) Justification

The Sobel operator is well suited here because it includes a built-in smoothing component within its kernel structure, giving it some resilience to noise while still being computationally simple and fast — an important balance for real-time feature-extraction pipelines. It also provides both gradient magnitude and direction, giving richer information for downstream feature descriptors than a basic Laplacian operator, which is direction-agnostic and more noise-sensitive.

(e) Presentation Summary

Reliable feature extraction requires enhanced, well-defined edges; the Sobel operator provides an efficient, direction-aware, moderately noise-resistant solution for this purpose.